

# Virtual Equipment System: Face Mask and Voodoo Doll for User Privacy and Self-Expression Options in Virtual Reality

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## ABSTRACT

Current trends in immersive technologies suggest an increase in capturing user's data to drive interactions and avatar representations. With growing numbers of data types being collected, users need an easy way to view and control their privacy settings. In this demo, we present a method for users to adjust options related to privacy settings, user data collection, and self-expression through the use of 3D user interface metaphors such as a mask and a voodoo doll.

**Index Terms:** Human-centered computing—Virtual Reality—;—Human-Computer Interaction—;

## 1 INTRODUCTION

Current trends in immersive technologies allow devices to capture more user data than before. New virtual reality headsets such as the HP Reverb G2 Omnicept and DecaGear are said to perform heart rate monitoring, pupillometry, eye-tracking, and facial expression tracking for the development of affective XR experiences.

Novel approaches and integrative methods are enabling wearables to track the user's physiology and emotional states. For example, headphones can track facial expressions using only miniature cameras near the ear to observe facial contour [3]. Oculus Reality Labs are working on technology that can track the user's hands through electromyography and machine learning.

While many of these advancements in data collection and tracking promise to improve the user's experience, they may also negatively impact the user's security. Fingerprints can be collected from a photograph [2]. Reflections in pupils can reveal the subject's location [1]. In addition, as the use of videoconferencing increases during the COVID-19 pandemic, there are reports of people being punished for objects inadvertently shown in the video. As such, organizations such as the Electronic Frontier Foundation and UC Berkeley's Office of Ethics recommend the use of virtual backgrounds on Zoom and other similar video conferencing software to protect users' privacy and security.

Audio and video are not the only forms of data gathered passively from the user. In the classic mouse and keyboard interface, users intentionally generate new data. They have control over when and what they generate through mouse clicks and keystrokes. In contrast, technologies such as voice, video, and motion data are a source of input data that is continuously tracked, whether or not the user intends on generating new data. Research has shown that motion controller data can identify 95% of the users viewing 360 VR videos with less than 5 minutes of tracking data per person. [4].

As we move into virtual and augmented worlds, even more of the user's data, behaviors, and emotional states will likely be continuously collected in sophisticated ways and without the user's continuous and explicit permission. Users need an intuitive interface to view and control their privacy settings in an immediate way.

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## 2 FACE MASK, VOODOO DOLL, AND MORE

In this demo, we extend our previous work on a Virtual Equipment System [8] and add a new piece of virtual equipment - a virtual face mask. We use the face mask as a metaphor for privacy. The mask symbolizes something that a user can use to conceal, protect, or show a different identity to the world. Previously, we have utilized our Virtual Equipment System to take advantage of the association between body parts and their corresponding senses [7]. For example, users can interact with Virtual Goggles or Virtual Headphones to adjust visual or audio settings, respectively.

Initially, we considered new privacy equipment that would replace headphones and goggles that would allow the user to adjust privacy options. However, privacy settings differ from audio and visual settings as they introduce a different context not directly associated with the senses and cover a much broader area of physical, physiological, and emotional layers. Settings affecting the experience of the user can be mostly addressed in visual, audio, and haptics terms; however, data generated by the users may also include heart rate, gestures and body poses, facial expressions, physical location, and many more.

To address the broader needs of privacy, we have developed a way for users to view and control the wide variety of data that they generate, drawing inspirations from the World in Miniature [6] and the Voodoo Doll paradigms [5]. These two techniques take objects of different scales and at different locations to place them in front of the user in easily interactable sizes. Whereas our inspirations utilize those metaphors for placing objects in a virtual world, we repurpose the doll metaphor for viewing and accessing privacy settings of the users themselves in our demo.

## 3 INTERFACE DETAILS

In our demo, the user is equipped with a transparent virtual face mask that does not block the user's vision when worn. The user can pull the face mask away from their face using 6 degree of freedom (6DOF) motion controllers. As the user pulls the mask away from their face, the mask becomes visible and morphs into a collection of 3D objects, including a voodoo doll, a background panel, and virtual cameras pointing at the voodoo doll (Figure 1). The user can interact with the voodoo doll and the background to adjust privacy settings. The user can also interact with the cameras to change how the privacy settings are applied to each of the different parties that utilize the user's data.

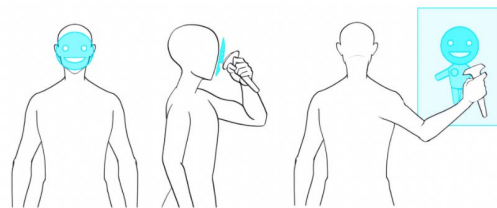


Figure 1: User grabbing a Virtual Face Mask to bring out the 3D menu based on the Voodoo Doll

### 3.1 Virtual Mask, Alt Node, and Equipment Gesture

The Virtual Mask is part of the Virtual Equipment System and supports that system's Alt Node and Equipment Gestures functionalities. Inspired by the alt key, an "Alt Node" is attached to the user's left controller and the user can move any Virtual Equipment to its proximity in order to access detailed menu options. In the case of the Virtual Mask, the user can interact with an Alt Node to bring up a 2D menu for privacy settings. Our system offers significantly more immediate and intuitive adjustment of privacy settings compared to the standard method, which requires the user to push the menu button and navigate through multiple text menus.

The user can also perform Equipment Gestures to the virtual mask. For example, the user can simply grab the mask and bring it down over the user's face. This mask down gesture symbolizes putting on a mask to enter into incognito mode where no user data is utilized or tracked. The user can do the opposite and pull the mask upward to perform an up equipment gesture. This gesture will exit incognito mode and restore the user's prior privacy settings.

### 3.2 Voodoo Doll

The voodoo doll represents the user and their associated privacy settings (Figure 2). Within our system, 3D spherical nodes are located on various parts of the voodoo doll. Each piece of hardware that collects user data based on a user's body is associated with a node on the corresponding body part of the voodoo doll. For example, the eye tracking hardware node is at the voodoo doll's eyes and the microphone node is at the mouth.

Using the 6DOF motion controllers, the user can interact with the nodes to enable or disable data collection in that area. Alternatively, the user can grab a node and move it to an "Alt Node" to bring up a detailed menu for the associated body part. In the case of the doll's eyes, the user can adjust privacy options such as eye tracking and pupil reflection as well as aesthetic expression options such as pupil color and pupil size.

### 3.3 Background Panel

The background panel is a 3D representation of incidental data collected by the system such as ambient noise and video of the user's surroundings. Through the background panel, the user can enable or disable background image or background noise. For detailed options, the user can bring the background to the "Alt Node" to access the menu that allows applying filters, customizing the background image, and tuning the background noise.

### 3.4 Virtual Cameras

Virtual cameras represent different parties that have permission to access the user's data. In our demo, we showcase four different virtual cameras - Multiplayer, System, Data Collection, and Master. Multiplayer Camera represents what other users in the virtual space would see. System Camera represents systems that utilize user's data to drive interaction, such as showing different emoticons based on the user's facial expressions. Data Collection camera represents services that collect and send data to third parties. Master camera is the camera that overrides all other cameras' settings.

The user can hover over any virtual camera to see what data each camera has access to. The current camera will be highlighted in green as shown in Figure 2. The user can also pick up the camera and use the trigger button to quickly enable or disable all data going to that camera.

### 3.5 Additional Privacy Options

Any options that are not associated with previous categories are listed on a side menu. These additional options could include the user's GPS location, IP address, and calorie expenditure.

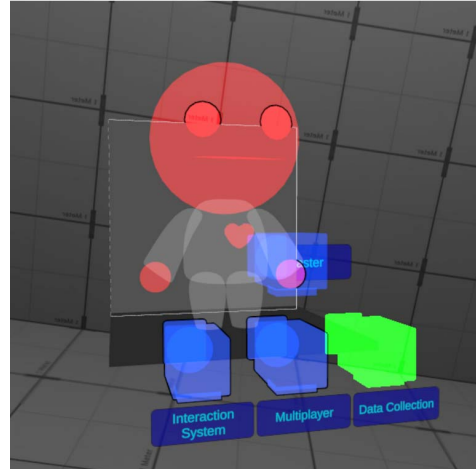


Figure 2: The Cameras representing different parties to the user's data. Data Collection camera is the active camera and is highlighted in green. The Voodoo Doll's parts are shown in red as the active camera lacks permission to access that data.

## 4 CONCLUSION

In this demo, we expanded upon our previous work of Virtual Equipment System by using metaphors such as face mask, voodoo dolls, and cameras. With the face mask equipment, the user can quickly access their privacy settings as well as perform a simple equipment gesture to enter incognito mode. With the voodoo doll, the user has a powerful visual aid to adjust privacy settings in fine detail. Lastly, with the virtual camera representing different observers, the user can easily visualize and manage which parties have access to the user's data. Together, these features provide users with an intuitive and powerful method to manage their privacy settings and self-expression.

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